

PROJECT TITLE

Project Title – ELECTRIC SELF DEFENCE SHOE USING ARDUINO

GROUP MEMBERS

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ABSTRACT

In today's world, personal safety—especially for women—is a critical concern. The proposed project introduces a wearable self-defense system embedded into footwear that serves two major purposes:

1. Physically deters an attacker through a safe electric shock.
2. Sends an emergency SMS alert to a trusted contact using GSM communication.

This shoe functions through a hidden trigger mechanism that activates a shock delivery system and an automated alert system. The integration of safety and communication in a single wearable device makes it a practical and innovative solution. Its low cost and portability allow it to be used by schoolgirls, working women, and elderly women alike.

OBJECTIVE

The core objective is to design and develop an **Arduino-based self-defense wearable device** that:

- Empowers women with **instant response capability** during emergencies.
- Integrates **electronics safely into everyday footwear**.
- Enables **wireless communication** through GSM modules for remote alerts.
- Ensures the solution is **affordable, durable, and easy to use**.
- Promotes **confidence, independence, and security** for women and girls.

COMPONENTS

1. Hardware Components:
 - Arduino Uno/Nano .
 - GSM Module (SIM800L/SIM900A)
 - High Voltage Generator (Boost Module)
 - Relay Module
 - Panic Button
 - Metal Plate
 - Rechargeable Battery
 - SIM Card
 - Wires, PCB, Shoe

2. Software Components:

- **Arduino Program (Firmware)** – Code written in Arduino IDE (C/C++)

COMPONENTS FUNCTIONALITY

1. Hardware Components Functionality:

- **Arduino Uno/Nano** – Acts as the brain of the system; runs the programmed logic, reads inputs (panic button), and controls outputs (relay, GSM module).
- **GSM Module (SIM800L/SIM900A)** – Communicates with mobile networks via the SIM card; sends emergency SMS to pre-stored numbers when triggered.
- **High Voltage Generator (Boost Module)** – Steps up voltage to produce a strong electric shock for self-defense.
- **Relay Module** – Works like an electronically controlled switch; Arduino turns it ON/OFF to safely control the high-voltage generator.
- **Panic Button** – A hidden emergency button; pressing it signals Arduino to activate the shock system and send alerts.
- **Metal Plate** – Conductive plate that delivers the electric shock when touched by an attacker.
- **Rechargeable Battery** – Portable power source for the entire circuit; ensures the system works even without external power.
- **SIM Card** – Stores network credentials and enables GSM module to send messages.
- **Wires, PCB, Shoe** – Provide electrical connections, mount the circuit, and conceal the system inside footwear.

2. Software Components Functionality:

Arduino Code (Firmware) – The set of programmed instructions that:

- Monitors the panic button state.
- Activates the relay to power the high-voltage generator.
- Sends SMS alerts through the GSM module.
- Can include delay timers, safety checks, and multiple recipient alerts

WORKING OF SYSTEM

The project functions based on a combination of **hardware and software triggers** controlled by Arduino.

Steps:

1. A hidden **panic button** inside the shoe is pressed by the user discreetly.
2. The **Arduino detects** this input signal and:
 - Activates a **relay-controlled high-voltage circuit**.
 - Sends power to the **shock module**, which delivers a brief electric pulse through metal electrodes placed in the toe or sole area.

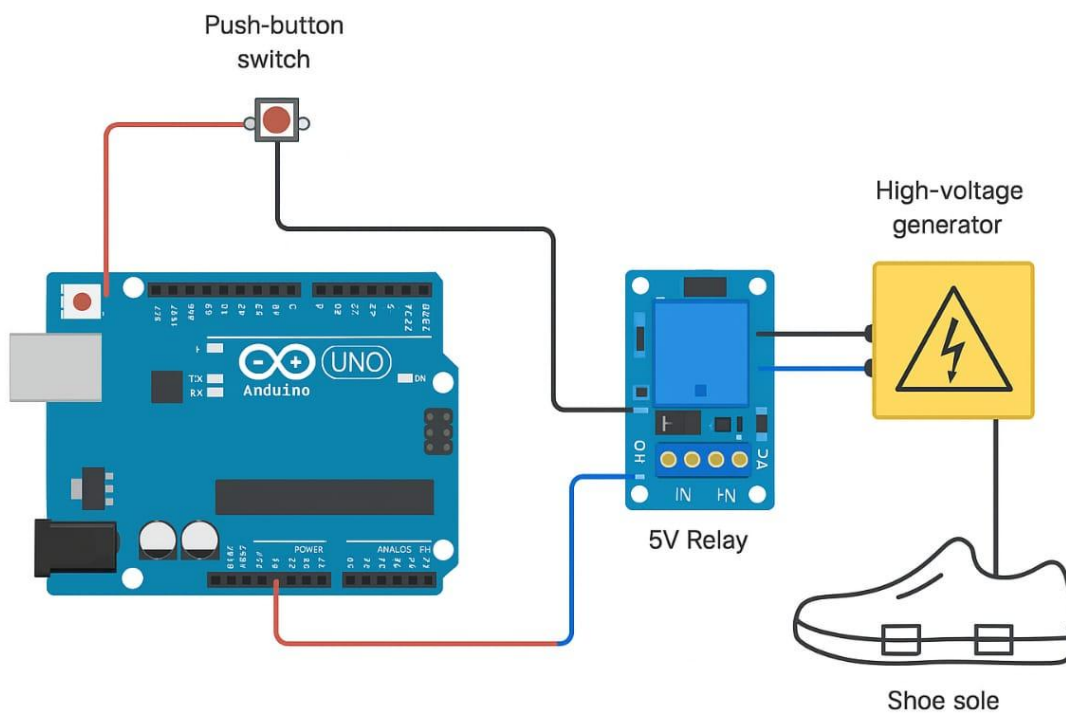
FUNCTIONALITY OF SYSTEM

System Name: Electric Self-Defense Footwear using Arduino Uno

Core Functionality:

1. Detection (Manual Trigger):
 - The user (girl) presses a small, hidden button embedded in the shoe or slide when she feels threatened.
2. Signal Processing:
 - The button is connected to an Arduino Uno.
 - When the button is pressed, Arduino receives a HIGH or LOW digital signal (depending on wiring).
3. Activation:
 - Arduino activates a relay module for a short duration (e.g., 3–5 seconds).
 - The relay allows power from the battery to flow to a high-voltage module (e.g., a 400kV step-up transformer or stun gun circuit).
4. Defense Action:
 - The high-voltage module charges and sends current to conductive metal pads or electrodes embedded in the sole.
 - When an attacker touches or grabs the wearer (or the shoe), a non-lethal electric shock is delivered, causing pain or surprise to allow escape.
5. Deactivation:
 - After a few seconds, Arduino turns off the relay.
 - The shock circuit is disconnected automatically to prevent overuse or accidental harm.

Circuit Diagram



THANK YOU!